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- **Semester: 3**
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- **Secondary Language for the BSP:** German

**Title:**

"The Effect of Observation Frequency on Entropy Estimation for a Gaussian Random Process"


**Abstract:**

This project investigates how the frequency of recorded observations affects entropy estimation for a temporally dependent random process. The student will use a one-dimensional Gaussian autoregressive model with a known marginal distribution and entropy. This controlled setting allows changes in estimation error to be studied while the true entropy remains constant.

The student will first explain why the model preserves its Gaussian marginal distribution and implement reproducible simulations in Python. Using a nearest-neighbour entropy-estimator formula supplied by the supervisor, the student will validate the implementation on small examples and independent Gaussian samples. The mathematical work will use elementary probability and properties of normal random variables.

The experiments will compare dependence strengths and recording intervals with either a fixed number of retained observations or a fixed simulation length. Independent Gaussian samples of matching sizes will provide reference controls. Across independent trajectories, the student will measure mean error, root mean squared error and simulation variability to assess the benefits and costs of less frequent recording.

The project will produce a concise report, comparative figures and reproducible code. It will provide a controlled benchmark for the wider study of observation effects in entropy estimation.